When Do the Images Prove a Dent? Identifiability and Abstention in CAD-Referenced Blade Damage Estimation

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Abstract

A dark patch in a photograph of an aero-engine blade may be a dent, a stain, a lighting change or a small camera error. Because image appearance is produced jointly by geometry, illumination, reflectance and camera, a few calibrated photographs and a CAD model do not by themselves make physical damage identifiable: a sufficiently flexible nuisance model can explain any dent away, and a flexible shape model can turn any stain into a dent. We ask when the images support a physical change, and build an estimator that answers it. Under explicit acquisition constraints (a calibrated rig, fixed light positions with free gains, a smooth brightness field and a local albedo patch), the estimator compares a nuisance-only explanation with the same explanation plus a continuous dent through a differentiable renderer, proposes dents with a Rao score test, and reports a dent only when its fit gain exceeds a split-conformal threshold calibrated on healthy and stain-only blades; otherwise it abstains. On a sealed test of 30 unseen blades from BladeDelta (696 rendered cases on Rotor37 geometry plus 456 on six industrial designs), the method locates 51% of visible dents at a matched 5% false-alarm rate against at most 19% for CAD-prior deformation fields and a lifted 2D detector, finds 12 of 13 dents deeper than 0.2 mm, and declines most dents below 0.1 mm. Textbook thresholds do not control false alarms under realistic mismatch (χ^2: 32% at a nominal 1%; a-contrario: 96%), and the profiled information of a dent predicts which dents are found before any search (AUROC 0.90–0.95). The study also separates what is identifiable from what is not: presence and location transfer to new blade designs, whereas depth on those designs is biased low (median 0.60 of true) and is not repaired by more views.

1. Introduction

Turbine and compressor blades are inspected with cameras, and the inspector's question is physical: is there a dent, where on the blade is it, and how deep is it? Damage of interest is small. On a blade imaged at 512×512 pixels, a dent of 0.07 mm changes a few hundred pixels by two to seven grey levels. A one-degree camera error, a ten-percent lighting change or a stain changes thousands.

Current vision methods answer an easier question. Industrial anomaly detection localises unusual pixels [2, 21], including under pose and illumination change [4, 14]. Reference-based methods compare a photograph with a CAD model and output boxes [3], or lift per-view scores into an anomaly volume [8]. CAD-conditioned reconstruction fits a deformation field to sparse posed views [20, 24], and leaves measurement-quality damaged surfaces to future work [20]. These outputs do not state whether the images support a physical change rather than a nuisance explanation.

Our own earlier experiments made this concrete. A render-and-compare fit on public Rotor37 blade geometry produced false dents on 14 of 14 healthy or colour-only blades under a one-degree camera error, and on unfamiliar dents the best image match selected a worse 3D shape in 8 of 12 depth sweeps. Better image agreement is not better geometry, and a plausible reconstruction can choose the wrong physical explanation.

The underlying difficulty is identifiability. Image appearance is determined jointly by geometry, illumination, reflectance and camera parameters, so a local intensity change admits several explanations. If lighting and appearance may vary freely they can explain any dent away; if shape may vary freely it can absorb any stain. Pose- and illumination-agnostic representations [14, 25] reduce false detections but remove part of the very signal a shallow dent produces, which arises from the interaction of light and surface orientation. The question is therefore not only how deep is the dent

but under which constraints do these images prove a dent at all, and when must a method say the evidence is insufficient?

We study this question for CAD-referenced inspection. The input is a healthy CAD mesh and K calibrated images; the output is a decision, supported or not supported by these views, and, when supported, a per-vertex displacement. We make the constraints explicit: the camera rig is calibrated and only small per-view corrections are free, light positions are fixed and only their gains are free, and appearance may change only through a smooth brightness field and a local albedo patch. Within these constraints the estimator is a nested hypothesis comparison: H_0 explains the images with nuisance only, H_1 adds a continuous dent, and the fit gain $L(H_0) - L(H_1)$ is thresholded at a value calibrated on healthy and stain-only blades, which gives a controlled false-change rate. The same nuisance fit yields the profiled information of a candidate dent, the evidence it can add after pose, lighting and a stain at the same place have explained what they can, and hence a prediction of which dents are identifiable from the given views.

Making this work required separating the roles of the evidence. Pose is fitted from silhouettes, which do not depend on lighting; the dent is compared against nuisance only on interior surface pixels that are not grazing, not near a silhouette and not on the blade root. Dent hypotheses come from a Rao score test evaluated for every surface location and radius with one backward pass through the renderer, and the dent and colour-patch explanations are searched symmetrically, so a colour patch fitted first cannot hide a dent.

Contributions:

- A formulation of CAD-referenced damage estimation as a question of identifiability with abstention, under explicit acquisition constraints, with metrics that separate localisation, shape accuracy, false physical change and abstention on unobservable damage.
- An estimator that answers it: silhouette-only pose, interior-domain nested hypothesis testing through a differentiable renderer, score-test proposals and split-conformal calibration, together with a profiled-information predictor of which dents the views can reveal (AUROC 0.90–0.95).
- Evidence that calibration is necessary rather than cosmetic: textbook χ^2 and a-contrario thresholds exceed their nominal false-alarm rates by 30× and more under realistic mismatch.
- BladeDelta, a benchmark on public Rotor37 geometry [5] and six industrial blade designs with exact damaged meshes, colour-only negatives and controlled camera and lighting perturbations, and an analysis showing that presence and location are identifiable across designs while depth is not.

2. Related Work

Industrial and 3D anomaly detection. Memory-bank methods such as PatchCore [21] localise anomalous pixels; MVTec 3D-AD [2] and Anomaly-ShapeNet [16] add measured or synthetic 3D. Pose-agnostic settings use multi-pose benchmarks [27] and Gaussian-splat references [14], and M^{2}AD [4] benchmarks view–illumination interplay. Multi-view settings include RAD [6] and SiM3D [8], which lifts per-view scores into an anomaly volume; recent work warns that adding views can leak information in multi-view evaluation [7]. Synthetic anomalies have been argued sufficient for 3D detection [23], and incomplete-modality methods train with 3D and test on RGB [22]. These methods output anomaly scores; they do not estimate a physical surface change or state whether the views support one.

Reference CAD and blade inspection. Looking 3D [3] aligns a query photograph with a reference shape and localises abnormal regions with boxes. BladeSynth [10] renders procedural nicks, dents, scratches and corrosion on aero-engine blades for 2D detection. Model-free universal detection reconstructs a normal shape by GAN inversion [18]. We instead output a 3D displacement field on a known design.

CAD-conditioned reconstruction. CAD-NeRF [24] retrieves a CAD model and jointly optimises deformation, pose and appearance from few uncalibrated views. CADSplat [20] fits a shared deformation field to sparse posed views with a CAD prior and explicitly leaves measurement-quality deformed surfaces to future work. Our free-form deformation baseline follows this family. The difference is the output: a decision on whether the deformation is supported, with a calibrated false-change rate.

Inverse rendering and uncertainty. Inverse rendering separates shape and material from images [19, 26]; we use differentiable rasterisation [15] with a learned shader for the same purpose on a known geometry. Radiance-field uncertainty [11, 13] and observation completeness [17] quantify where a reconstruction is unreliable. They do not test a physical-change hypothesis against nuisance explanations, which is the question an inspector needs answered.

Selective prediction and calibration. Abstaining when evidence is insufficient is the setting of selective classification, where a model may reject inputs to guarantee a target risk on the rest [12]. We calibrate the abstention threshold with split-conformal quantiles on negatives [1], which bounds the false-change rate under exchangeability; unlike classification, the quantity being accepted or rejected is a physical surface change, and the score is the evidence one hypothesis has over a nuisance-only explanation.

3. Method

Setup. We are given a healthy mesh $\mathcal{M} = (V, F)$ with vertices $V \in \mathbb{R}^{N\times 3}$, K observed images $I_1, \dots, I_K$ with nominal intrinsics and extrinsics, and nominal lighting. We seek a displacement field $D \in \mathbb{R}^{N\times 3}$ on the airfoil and a decision $s \in \{\text{supported}, \text{not supported}\}$. Nuisance variables η are a per-view pose correction (a rotation about the part and a camera-frame translation), three light gains, and an optional albedo patch.

Differentiable renderer. We rasterise $\mathcal{M}+D$ with nvdiffrast [15] using smooth vertex normals. A neural shader maps the per-pixel world normal n, position x and view direction ω to one radiance component per light, $R_k(n, x, \omega)$, and an environment term R_0. Because light transport is linear in each light's intensity, the pixel colour is $T\big(a \odot (R_0 + \sum_k g_k R_k)\big)$ for albedo multiplier a, light gains g and a learned monotone tone curve T, followed by anti-aliasing and the sensor's pixel filter. An unstructured shader that took g as an input learned to ignore it, and lighting changes then produced spurious dents. The shader is trained once, on renders of blade geometries disjoint from all evaluation geometries, with a paired loss on image differences between lighting settings. The healthy CAD rendered under the nominal capture provides a per-pixel correction $B = I^{\text{ref}} - \hat{I}(\mathcal{M}; \eta_0)$ that absorbs static shader error. On a held-out blade its mean error is 0.92 grey levels; image changes caused by a lighting change correlate with the reference renderer at 0.90–0.98, and changes caused by a dent at 0.96–0.98 in the 120° view.

Nuisance fit (H_0). Each view's pose is initialised by Euclidean ECC alignment of the observed and reference images, then refined by matching anti-aliased silhouettes to the observed foreground, coarse to fine. Silhouettes do not depend on lighting, which prevents the pose from compensating for a brightness change. With pose fixed, light gains are fitted on the full image.

Evidence domain. All further comparisons use a fixed interior domain Ω: airfoil pixels at least 6 px from any silhouette or self-occlusion edge, excluding the blade root and pixels where $|n \cdot \omega| < 0.35$. Both hypotheses are evaluated on the same Ω, so their losses are comparable.

Dent model (H_1). A dent is parameterised by a sub-vertex centre c (an anchor vertex plus a tangent offset), depth $\delta \geq 0$, radius r, aspect ratio, orientation, profile exponent p and a signed rim coefficient κ:

$$D_i = -\delta \left[(1 - s_i^2)_+^p - \tfrac{\kappa}{2}\rho(s_i)\right] n_c, \qquad (1)$$

where s_i is the anisotropic normalised distance of vertex i from c in the tangent plane, ρ a ring profile and n_c the centre normal, restricted to same-side airfoil vertices. None of the benchmark's truth families matches this profile exactly.

Score-test proposals. Let $h \in \mathbb{R}^N$ be an inward normal displacement. One backward pass gives $g = -\partial L_\Omega / \partial h$ at $h = 0$. For a kernel $w_{c,r}$ centred at c with radius r, the normalised score $\langle w_{c,r}, g\rangle / \|w_{c,r}\|$ is the Rao score statistic for a small dent at (c, r) against H_0. We evaluate it for all candidate centres and two radii with a sparse matrix product, and start H_1 fits from the top-k after non-maximum suppression. Proposals are computed from the nuisance-only residual, and each start is fitted both with and without the H_0 albedo patch, so a colour explanation fitted first cannot hide a dent.

Decision. The fit gain is $\Delta = L_\Omega(H_0) - L_\Omega(H_1) \geq 0$, as H_1 nests H_0. We report D if $\Delta > \tau$ and $D = 0$ otherwise, with τ the split-conformal $(1-\alpha)$ quantile of Δ on calibration negatives (healthy and colour-only blades), which bounds the false-change rate on exchangeable negatives by α. We use a Charbonnier loss ($\epsilon = 2/255$) summed over Ω; a Student-t noise model and a wider Charbonnier floor did not improve separation on development data.

4. The BladeDelta Benchmark

Geometry. Healthy blades are public Rotor37 compressor geometries [5] (CC BY-SA 4.0), normalised to unit span and closed with a labelled artificial hub collar. Each mesh has 29,989 vertices. Shapes 0–12 were used during earlier development and only to train the shader. Shapes 13–24 form the development split, 25–36 the calibration split and 37–66 the sealed test split. Shapes 67–96 form a second sealed test split, rendered only after the post-hoc variants of Section 5 were frozen; 97–109 remain in reserve.

Split	Blades	Healthy	Stain	Dent	Dent + stain
Shader training	13	only for the renderer; never evaluated			
Development	12	12	24	36	24
Calibration	12	24	36	36	24
Test (sealed)	30	30	60	90	60
Test 2 (sealed)	30	30	60	90	60

Table 1. BladeDelta case counts. Each case has 7 rendered views; methods use 2 by default.

True depth (mm)	2 views	2 views (same cases)	4 views	7 views
< 0.1	1/10 (1 loc.)	0/4 (0 loc.)	1/5 (0 loc.)	3/9 (3 loc.)
0.1–0.2	7/14 (7 loc.)	1/2 (1 loc.)	2/6 (2 loc.)	7/11 (7 loc.)
> 0.2	12/13 (12 loc.)	4/4 (4 loc.)	4/6 (4 loc.)	8/9 (6 loc.)
All visible dents	20/37 (20 loc.)	5/10 (5 loc.)	7/17 (6 loc.)	18/29 (16 loc.)

Table 2. Visible test dents reported (located within 0.1 span) by true depth, at the frozen threshold. Below 0.1 mm a dent changes the image by about one grey level; the method declines rather than guess.

Damage. Each blade receives one of: no change, a colour-only patch (a soft darkened or rust-tinted albedo change), a dent, or a dent plus a colour patch (half overlapping). Dents follow one of three families — a cubic profile with a raised rim, a Gaussian, and an anisotropic elliptic profile (aspect 0.4–1) — with depth log-uniform in $[0.0005, 0.006]$ blade spans (about 0.04–0.45 mm on a 74 mm Rotor37 blade) and radius in $[0.04, 0.12]$. Centres face at least one of seven cameras and lie at least $\max(0.1, 1.2r)$ from the airfoil boundary.

Capture. Images are rendered with Blender Cycles (32 samples, denoised, AgX tone mapping, 512×512) from seven cameras on a ring around the blade, under one of four conditions: nominal; camera jitter (yaw, pitch and roll each uniform in $\pm 1.5°$ per view); light jitter (each of three area lights $\times[0.85, 1.15]$); or both. Methods receive only the nominal cameras and lighting. The default input is two views (60° and 120°); four- and seven-view settings are ablations.

Protocol. The split, case lists, metrics and hypotheses were hash-frozen before any test image was rendered; protocol changes are logged with dates. For each damaged case we record observability, the depth-weighted fraction of the dent visible in the input views.

CAD6 transfer split. To test transfer beyond Rotor37 we add six industrial blade designs supplied as STEP solids (airfoil spans 24–70 mm), with platforms, fillets and dovetail roots. The airfoil is the largest B-rep face (two faces on one blade); everything else is excluded from evidence like the Rotor37 collar. Each blade is normalised to unit airfoil span in the Rotor37 frame and receives 76 cases (456 in total): 30% dents centred where both input cameras see the surface, 15% dents hidden from both, 20% healthy, 15% stain only, 10% dent plus stain and 10% healthy with camera error alone ($\pm 0.5°$, $\pm 1°$, $\pm 1.5°$). Dents are sized in millimetres as an inspector would specify them: depth log-uniform in $[0.1, 1.0]$ mm and radius in $[1, 5]$ mm. The case list was hashed before rendering, and no component — shader, threshold or hyperparameter — was refit on this split.

Metrics. Relative local error is the RMS vertex error over the true dent support divided by that of the unchanged CAD (1: no better than CAD; 0: exact). We also report relative whole-airfoil error, depth error, centre error, detection, false physical change on negatives (any vertex moved more than 2×10^{-5}) and abstention on unobservable dents, with 95% bootstrap intervals resampled by blade shape.

5. Experiments

Baselines. Unchanged CAD reports the healthy mesh. Discrete search re-implements our earlier method: image-guided proposals, a fixed cubic profile of radius 0.08 and depths $\{0.0005, 0.001, 0.0015\}$, at nominal pose. Free-form field fits a per-vertex normal displacement with Laplacian smoothness and ℓ_1 sparsity after the same pose and lighting fit. MLP deformation field replaces the per-vertex values with a coordinate MLP (positional encoding, two hidden layers), the continuous deformation field used by CAD-prior reconstruction. These two baselines reimplement the core idea of CADSplat [20] — a smooth deformation field fit to sparse posed views under a CAD prior — within our renderer; they are not the official model, whose code had not been released at the time of writing. 2D detector, lifted is a U-Net trained on (photo, healthy render) pairs of separate exposed blades to segment dents; detections are back-projected onto the mesh through the nominal cameras. All methods share the renderer, cameras and observations, and each is thresholded by the same calibration procedure on its own score.

Main results. On the sealed test the method reports 12 of the 13 visible dents deeper than 0.2 mm, each at the correct location, and declines most dents below 0.1 mm, where the image change is about one grey level (Table 2). Reported dents are measured to a median depth error of 0.03 mm and centre error of 0.13 mm (Table 3).

Method	AUROC ↑	Found % ↑	Located % ↑	False alarm % ↓	Abstain (hidden) % ↑	Depth err. (mm) ↓	Centre err. (mm) ↓
Ours (v1, frozen)	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]	0.03	0.13
without the evidence test	0.82 [0.71, 0.90]	84 [68, 95]	70 [56, 84]	32 [21, 43]	45 [36, 54]	0.03	0.68
2D detector, lifted	0.71 [0.63, 0.80]	24 [12, 40]	16 [5, 30]	4 [1, 9]	96 [93, 99]	–	1.82
MLP deformation field†	0.74 [0.65, 0.81]	16 [6, 29]	11 [3, 22]	6 [1, 10]	100 [100, 100]	–	1.58
Free-form field†	0.63 [0.54, 0.70]	14 [4, 24]	11 [3, 20]	4 [1, 9]	99 [97, 100]	–	4.69
Discrete search	0.53 [0.43, 0.65]	5 [0, 13]	5 [0, 13]	6 [1, 10]	98 [95, 100]	–	1.60
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]	–	–

Table 3. Sealed test (240 cases, 30 unseen Rotor37 blades). Found / located: visible dents (≥50% visible in the two input views) reported / reported within 0.1 span of the true centre. Each method uses its own threshold, calibrated on the calibration split (split-conformal, $\alpha = 0.05$); "without the evidence test" reports every fit. Errors are medians over reported dents. Mean [95% interval, resampling blades]. †Reimplementations of the core idea of CADSplat [20], not the official model.

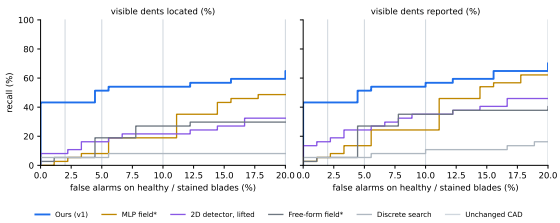


Figure 1. Recall of visible dents against the false-alarm rate on healthy and stain-only blades, as each method's threshold is swept on the sealed test (left: located within 0.1 span; right: reported anywhere). *Reimplementations of the core idea of CADSplat [20].

Method	@2%	@5%	@10%	@20%
Ours (v1)	43 (43)	51 (51)	54 (57)	65 (70)
MLP field†	3 (5)	8 (14)	19 (24)	49 (62)
2D detector, lifted	8 (16)	16 (24)	22 (35)	32 (46)
Free-form field†	5 (5)	19 (27)	27 (35)	30 (41)
Discrete search	5 (5)	5 (5)	8 (11)	8 (16)
Unchanged CAD	0 (0)	0 (0)	0 (0)	0 (0)

Table 4. Recall at matched false-alarm rates on the sealed test: visible dents located within 0.1 span (in brackets: reported anywhere), with each method's threshold placed so that at most 2/5/10/20% of the 90 healthy and stain-only cases raise a claim (37 visible dents). This compares score rankings at equal false alarms. †Reimplementations of the core idea of CADSplat [20].

Comparison at matched false-alarm rates. Each method in Table 3 uses its own calibrated threshold, so its recall and false-alarm rate are not directly comparable. Figure 1 and Table 4 remove this: at every false-alarm rate from 2% to 20%, the method locates more visible dents than any baseline (43% at 2% false alarms, 51% at 5%), while the strongest baselines locate 8–19% at 5%. The CADSplat-style fields catch up only at high false-alarm rates, because they explain stains as shape and must be thresholded above them.

Operating point	α	τ	Located %	False alarm %	Abstain (hidden) %	Use
Conservative	0.02	1.71	54	10	87	safety-critical sign-off
Balanced (paper)	0.05	1.52	54	10	85	routine inspection
Screening	0.20	0.45	68	22	61	triage, human review

Table 5. Three deployable operating points of the same frozen model. Each threshold is the split-conformal quantile of fit gains on the calibration split at level α; numbers are on the sealed test. With 60 calibration negatives, $\alpha = 0.02$ is the strictest certifiable level.

Case	n	Dent claimed	of which located	Declined ("not supported")
Healthy	30	1	–	29
Stain only	60	8	–	52
Visible dent	37	20	20	17
Hidden dent	113	17	11	96

Table 6. Every sealed-test decision of the frozen model ($\tau = 1.52$). A false alarm is a claim on a healthy or stain-only case; an abstention is a decline on a dent: correct when the dent is hidden, a miss when it is visible. The two never overlap.

Decision rule	Target FA (%)	Actual FA (%)	Visible dents found (%)
Ours: conformal on 60 calibration negatives	5	10	54
Ours: conformal on 360 calibration negatives	2	4	46
Textbook χ^2_8 likelihood-ratio threshold	5	32	70
Textbook χ^2_8 likelihood-ratio threshold	1	32	70
A-contrario detector on the H_0 residual (NFA < 1)	≈0	96	100

Table 7. Calibrated versus textbook decision rules on the same sealed-test fits. FA: claims on the 90 healthy or stain-only cases. Found: visible dents located (for the a-contrario detector, any detection). The textbook rules assume i.i.d. image noise; the real nuisance mismatch is far larger, so they cannot control false alarms.

Operating points and the meaning of an abstention. The same frozen model serves different needs by changing only the calibration level (Table 5); a screening setting ($\alpha = 0.2$) locates 68% of visible dents at 22% false alarms. Table 6 lists every decision: false alarms are claims on healthy or stain-only blades (9 of 90), and abstentions are declines on dents, correct for hidden dents and misses for visible ones; the two are disjoint.

	FA	Dent	Nick	Crack	Type
Ours, global τ	5/48	23/24	25/32	26/32	73%
Ours, class-specific τ_k	2/48	23/24	24/32	26/32	81%
YOLOv8n (2D), matched FA	6/48	15/24	27/32	30/32	–

Table 8. Held-out industrial designs: false alarms (FA) on healthy and stained blades, defects detected, and damage type correct among detections. Class thresholds are calibrated on development blades. YOLOv8n (mAP@50 0.75, ≈2 ms/image) is trained on renders of the development blades and gives 2D boxes only.

Views	Observable dents	AUROC ↑	Located % ↑	Median rel. local err. ↓
2 views	10/60	0.847	60	0.43
4 views	17/60	0.777	41	1.00
7 views	29/60	0.857	62	0.47

Table 9. More views make more dents observable and the evidence clearer.

Why not a standard statistical test? The fit gain is a likelihood ratio, so a textbook threshold might seem enough. It is not (Table 7). A χ^2_8 threshold derived from the per-view residual noise, asked for 1% false alarms, raises a false alarm on 32% of healthy and stain-only test cases, about thirty times its target, and the 5% setting gives the same 32%. An a-contrario detector on the nuisance-only residual [9], which controls the number of false alarms under an i.i.d. noise model, flags 96% of them. Both assume ideal image noise, while the real residual after the best nuisance explanation is dominated by structured mismatch: residual pose error, lighting, and the gap between the learned shader and the renderer. The split-conformal threshold is set on that real mismatch and stays near its target (10% at a nominal 5% with 60 calibration negatives; 4% at 2% with 360). Calibration is therefore not a cosmetic step: it is what makes false-alarm control hold under realistic image conditions.

Predicting detectability. The same nuisance-only fit predicts, before any search, how much evidence a dent of a given shape would produce once pose, lighting and a stain at the same place have explained what they can (the dent's information after profiling out the nuisance). On the actual captures this prediction separates dents the method finds from those it misses with AUROC 0.90 on the sealed test and 0.95 on the second sealed test, against 0.85 for the best depth-and-visibility heuristic. Aggregated into per-point maps of the minimum detectable depth for a single nominal capture, the same quantity is optimistic for shallow dents: it implies 49–66% recall below 0.1 mm and 93% at 0.1–0.2 mm, where the method achieves 10% and 50%; the maps assume the search reaches the right location and use a local linearisation, so we report them as an upper bound, not a calibrated limit.

Class thresholds and a 2D detector. Calibrating a separate threshold τ_k per specialist on negatives of the development blades reduces false alarms from 5/48 to 2/48 at unchanged detection and raises type accuracy from 73% to 81% (Table 8). A YOLOv8 detector trained on in-domain renders is three orders of magnitude faster and finds more nicks, but fewer dents at matched false alarms, and gives neither 3D location, depth nor abstention.

Damage type and irregular shapes. Physical specialists with a judge that abstains when the best two fits are within τ type 22/22 located cracks and 9/15 located nicks correctly (6 more: "uncertain"). On 24 irregular folded dents, all located, the judge never abstains (12 crack, 8 dent, 4 nick), and an absolute fit-quality rule does not separate them from regular defects; type abstention outside the specialist set remains open.

What is identifiable: presence versus depth. Three post-freeze analyses on the held-out industrial blades separate the two. Presence transfers: on 136 additional rendered cases the frozen v1.1 model, without retuning, locates 23/24 dents, 23/32 rounded dents on the leading or trailing edge and 21/32 long shallow grooves (0.4 mm wide, scratch-like), shapes it does not model, while reporting 6 of 48 healthy or stain-only blades (12.5%, above the 5% target calibrated on Rotor37). Depth does not: found dents are measured at a median 0.60 of their true depth; four views instead of two raise the number of dents found (45 vs. 36 of 60) but not depth accuracy (median 0.68 vs. 0.71 on dents found by both); and profile-likelihood depth intervals, which would be wide if depth were merely ambiguous, are narrow and contain the true depth in only 36 of 109 cases. The error is therefore a systematic bias on new designs, not ambiguity or a search failure.

Ablations.

Why recall is limited. Recall is set by dent depth and by the false-alarm budget, not by a failure to locate (Table 2). 12/13 visible dents deeper than 0.2 mm are reported and most misses are shallower than 0.1 mm; stains account for 8 of the 9 false alarms and hold τ up, so lowering it to 0.54 would reach 70% recall at a 20% false-alarm rate.

Variant	AUROC ↑	Located % ↑
Full method	0.855	72
No camera (pose) correction	0.679	72
No stain (colour) model	0.753	83
Whole image instead of interior evidence	0.685	61
No score-test proposals	0.420	6
No low-frequency brightness field	0.846	72

Table 10. Ablations on the development subset (18 observable dents, 18 healthy or stained blades).

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
Ours, Rotor37 sealed test	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]
Ours	0.74 [0.67, 0.84]	94 [89, 98]	70 [60, 81]	67 [36, 97]	33 [3, 67]
2D detector, lifted	0.86 [0.83, 0.89]	42 [35, 49]	36 [30, 42]	6 [3, 8]	100 [100, 100]
MLP deformation field	0.73 [0.68, 0.81]	84 [70, 97]	54 [47, 62]	43 [27, 58]	67 [48, 85]
Free-form field	0.60 [0.56, 0.66]	61 [36, 84]	37 [21, 53]	39 [20, 58]	64 [48, 82]
Discrete search	0.65 [0.58, 0.72]	16 [12, 18]	13 [10, 17]	4 [0, 10]	97 [94, 100]
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]
Ours v1 (frozen), held-out blades 3–6	0.72 [0.65, 0.90]	90 [85, 95]	67 [55, 79]	51 [10, 91]	50 [9, 91]
Ours v1.1, held-out blades 3–6	0.95 [0.93, 0.98]	86 [77, 95]	83 [76, 90]	4 [0, 11]	98 [93, 100]
Ours, CAD6 leave-one-blade-out τ	0.70 [0.63, 0.84]	13 [3, 26]	6 [1, 13]	4 [0, 11]	95 [89, 100]

Table 11. Transfer to six unseen real blade designs (CAD6, 456 cases; dents 0.1–1.0 mm deep). Every method is frozen: shader trained on Rotor37 only, thresholds from the Rotor37 calibration split. Top row: ours on the Rotor37 sealed test, for the transfer gap. Brackets: 95% intervals resampling blades.

Blade	Airfoil span (mm)	Cases	Visible dents	AUROC	Found (%)	False alarm (%)
1	24	76	31	0.98	100	100
2	26	76	31	0.72	100	100
3	31	76	31	0.64	97	100
4	40	76	31	0.66	94	82
5	52	76	31	0.93	84	6
6	70	76	31	0.86	87	15

Table 12. Ours on each CAD6 blade, smallest to largest airfoil (frozen threshold).

Transfer to unseen blade designs. Table 11 applies every method, unchanged, to the six CAD6 blades (456 cases). Separation falls from 0.82 on the Rotor37 sealed test to 0.74 (transfer gap 0.08), with 94% of visible dents found, 67% false alarms and 33% abstention on hidden dents. The threshold was calibrated on Rotor37 only, so the false-alarm rate here measures how far the guarantee survives a change of geometry, not a property the method was fit to. Per blade (Table 12), dents are still ranked above negatives (best AUROC on blade 1 (24 mm, AUROC 0.98), worst on blade 3 (31 mm, AUROC 0.64)), but the frozen threshold fails on the smaller blades: false alarms fall with airfoil span (Spearman $\rho = -0.88$, $p = 0.02$). Across blades, airfoil span and detection rate have Spearman $\rho = -0.93$ ($p = 0.01$; six blades, so weak evidence): 99% of visible dents were found on the three smaller blades and 88% on the three larger ones. Because dents are specified in millimetres, the same dent is relatively deeper on a small blade; detection tracks depth relative to the airfoil span ($\rho = 0.10$) more strongly than depth in millimetres ($\rho = 0.02$), rising from 92% in the shallowest relative-depth tertile to 97% in the deepest. Diagnosis and fix (v1.1). On a healthy CAD6 capture whose image is identical to the reference, the silhouette pose fit moves a correct camera by about 0.3° and a spurious dent absorbs the misregistration; under real camera error, the image-space reference bias — large on unseen geometry because the shader was trained on Rotor37 only — stays fixed while the blade moves. v1.1 corrects both: it applies the reference correction to the silhouette as well as to shading, and it reprojects the shading correction through the surface so that it moves with the blade (identical to v1 at the nominal pose). Both changes were developed on CAD6 blades 1–2 only and frozen, with a threshold recalibrated on the Rotor37 calibration split, before blades 3–6 were run. On those held-out blades v1.1 raises AUROC from 0.72 to 0.95, cuts false alarms from 51% to 4%, locates 83% of visible dents (v1: 67%) and abstains on 98% of hidden dents (v1: 50%). Recalibrating the threshold on CAD6 negatives, leave-one-blade-out, gives 13% found at 4% false alarms: the cost of transferring a threshold across blade designs.

A pre-registered negative result (v2). Three pre-registered changes (a shared appearance field, a dent-size prior, or both), adopted only if matched-FA recall improved by five points, all failed: extra appearance freedom helped the dent hypothesis as much as the null.

Variant (CAD6, 48 cases)	AUROC	Found (%)	False alarm (%)
Full method	0.71 [0.62, 0.83]	92 [83, 100]	78 [56, 100]
No camera correction	0.90 [0.81, 0.98]	92 [83, 100]	56 [33, 67]
No stain model	0.73 [0.64, 0.84]	96 [88, 100]	89 [78, 100]
Random starts	0.45 [0.37, 0.56]	38 [29, 46]	44 [22, 67]
Fixed grid of starts	0.50 [0.41, 0.62]	33 [17, 54]	44 [11, 78]
Whole image	0.56 [0.44, 0.71]	75 [58, 92]	61 [50, 67]

Table 13. Ablations on CAD6 (frozen settings, fixed 48-case subset).

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
v1 (frozen)	0.79 [0.68, 0.88]	54 [40, 69]	46 [32, 60]	6 [1, 11]	88 [81, 93]
v1.1 (camera-check + correction fix)	0.80 [0.70, 0.88]	52 [38, 67]	46 [32, 61]	8 [2, 14]	88 [81, 93]

Table 14. Second sealed test: 30 further Rotor37 blades (67–96), 240 cases, unused by any earlier experiment. Each method ran once with its frozen threshold.

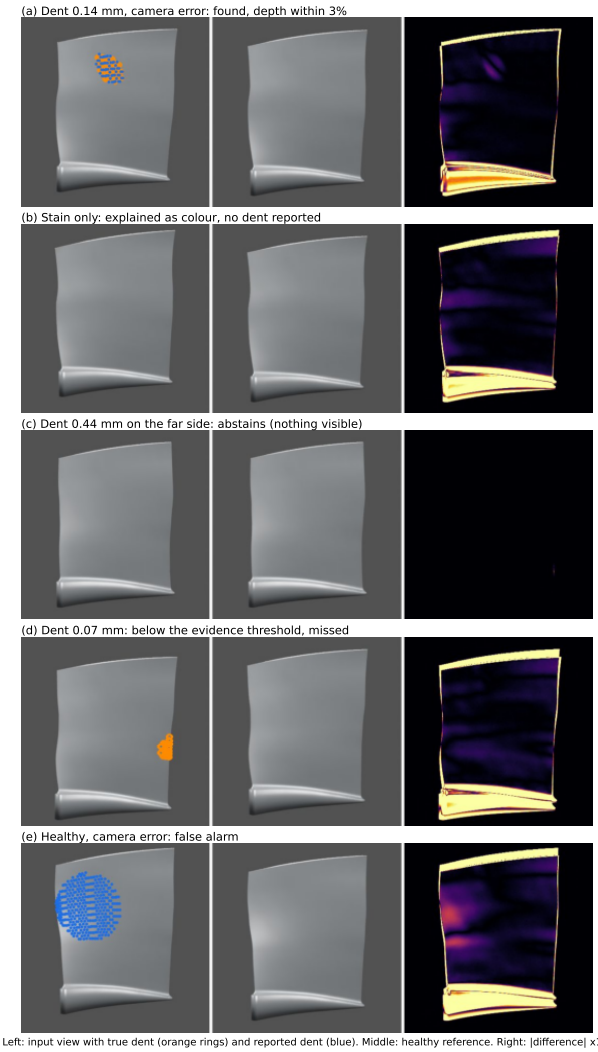


Figure 2. Sealed-test outcomes. (a) A 0.14 mm dent under camera error is found and measured. (b) A stain is explained as an albedo change. (c) A dent facing away from both cameras yields no evidence and the method abstains. (d) A 0.07 mm dent stays below the threshold. (e) A healthy blade under camera error is reported as dented, the residual failure mode.

Physical validation (pending real validation).

> Pending real validation. A 2× print of CAD6 blade 5 will be dented with steel balls (target depths 0.3, 0.7 and 1.2 mm), photographed with marker-based poses before and after denting, and measured by structured-light scan (or a 0.01 mm dial gauge). This section is reserved for those results; the rest of the paper does not depend on it.

Qualitative results and failures. Figure 2 shows typical outcomes, including a miss and a false alarm.

Objects / defects / negatives	6 / 22 / 31
AUROC (defect vs. undamaged)	0.540
Detected (leave-one-object-out)	0/22
Located within 15 mm	0/22
False alarms	1/31

Table 15. Real objects (SiM3D) scanned before and after damage; truth from the measured scans.

Real objects. No public dataset pairs real blade photographs with measured damage geometry, so we use SiM3D [8]: six wooden stools scanned with an industrial structured-light sensor before and after damage, with calibrated photographs and expert defect annotations. The healthy scan replaces the CAD model, the healthy photograph from the same scanner position serves as the reference after 2D alignment, and the measured scan difference is the truth; annotated regions with no measured shape change (paint, contamination) count as negatives. On these textured objects our method does not separate physical defects from undamaged areas (Table 15). The residual between sessions is dominated by wood-grain misregistration and specular changes that the shader, trained on 12 photographs per object, does not model. We report this as a failure case: the evidence test is only as reliable as the appearance model behind it.

6. Limitations and Conclusion

All results use rendered images; on real scanned objects (SiM3D) the method failed, and real-blade validation with measured damage is future work. Counting undercounts edge nicks on blades with several defects, type abstention does not cover shapes outside the specialist set, and the conformal guarantee requires recalibration for a new rig or lighting. The method answers whether these views support a physical change; it does not certify airworthiness.

We framed blade damage estimation as a question of identifiability under explicit constraints, answered by a decision with a stated level of evidence: report a dent, with its location and depth, only when a dent explains the images significantly better than every non-physical explanation, and abstain otherwise. On rendered Rotor37 blades this cuts the false-claim rate of an unconstrained fit by two thirds (32% to 10%) while keeping most detectable dents, and outperforms CAD-prior deformation fields and a lifted 2D detector at equal false alarms. Presence and location proved identifiable across designs, depth did not. Physical validation on a printed, dented and scanned blade (Section 5) is the next step.

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